California Regional Water Quality Control Board

North Coast Region

ORDER NO. 94-71
ID NO. 1A89001NDN

WASTE DISCHARGE REQUIREMENTS

FOR THE

CALIFORNIA DEPARTMENT OF CORRECTIONS
PELICAN BAY STATE PRISON

Del Norte County

The California Regional Water Quality Control Board, North Coast Region (the RWQCB), finds that:

1. The State of California, Department of Corrections (the permittee) submitted a Report of Waste Discharge, dated November 12, 1993 for the Pelican Bay State Prison, and applied for renewal of its Permit to discharge treated wastewater under the National Pollutant Discharge Elimination System (NPDES).

2. The Prison's wastewater treatment plant is designed to treat an average 0.75 million gallons per day (mgd) of sewage to an advanced level which results in pathogen-free and nutrient-controlled effluent. The facility consists of a headworks with a coarse screen and grit removal, extended aeration - activated sludge (oxidation ditches), clarification, coagulation/filtration, chlorination/dechlorination and effluent disposal at a series of Rapid Infiltration Basins (RIBs).

3. Solids disposal consists of sludge storage in surface-aerated falcultative lagoons during the wet season. During the dry summer period, sludge is injected on 32 acres of land located within the prison grounds. The application rate is managed to balance nitrogen input to uptake by grass-crop harvesting.

4. Following dechlorination, treated effluent is discharged to 8.4 acres of infiltration basins located on a 26-acre parcel approximately two miles northeast of the prison and within the 100-year floodplain of the Smith River. The basins are surrounded by berms with a top elevation one foot higher than the standard project (1000-year) flood elevation..

5. A small quantity of treated and disinfected wastewater is used for landscape irrigation of the treatment plant grounds. The irrigation is limited to an area immediately south of treatment plant and east of the north detention basin. Other areas within the prison grounds may also be irrigated using effluent but have not, as yet, been identified. The effluent quality is more than sufficient to conform to all Wastewater Reclamation Requirements for this type of use.

6. The NPDES program requires permits for any "point source" discharge into waters of the United States. 40 CFR 122.2 defines "point source" as ..."any discernible confined, and discrete **conveyance** (emphasis added), including but not limited to , any pipe, ditch channel, tunnel, conduit, well, discrete fissure, container, rolling stock, concentrated animal feeding operation, landfill leachate collection system, vessel or other floating craft from which pollutants are or may be discharged..." The RIB's do not fit this definition. Consequently, an NPDES Permit is not being issued. The discharge is being regulated under the Porter Cologne Water Quality Control Act in the form of Waste Discharge Requirements.

7. The RWQCB adopted Water Quality Control Plans for the Klamath River Basin (1A) and the North Coastal Basin (1B) on March 20, 1975. The Klamath River Basin Plan (1A) was combined with the North Coastal Basin Plan (1B) to form the Water Quality Control Plan for the North Coast Region. The Plan for the North Coast Region was adopted by the RWQCB on April 28, 1988 and approved by the State Water Resources Control Board (SWRCB) on November 15, 1988. The Plan includes water quality objectives, implementation plans for point source and nonpoint source discharges and statewide plans and policies.

8. The beneficial uses of the Smith River include:

   a. municipal and domestic supply
   b. agricultural supply
   c. industrial service and process supply
   d. groundwater recharge
   e. freshwater replenishment
   f. water contact recreation
   g. non-water contact recreation
   h. ocean commercial and sport fishing
   i. warm freshwater habitat
   j. cold freshwater habitat
   k. wildlife habitat
   l. fish migration
   m. fish spawning
   n. shellfish spawning

9. Beneficial uses of areal groundwaters include:

   a. domestic water supply
   b. agricultural water supply

10. The Smith River is a component of the California Wild and Scenic Rivers System due to its extraordinary scenic, recreation, fishery, and wildlife values (The Smith River is also a component of the Federal Wild and Scenic Rivers System). The reach between the confluence of the Middle Fork and South Fork to the ocean is classified as "recreational". The Public Resources Code requires all state agencies to exercise their powers in a manner consistent with the policy and provisions of the Wild and Scenic Rivers Act. The Smith River is also a designated National Recreational Area (1991) under the U.S. Department of Agriculture (U.S. Forest Service).

11. The Water Quality Control Plan includes water quality objectives which are not to be exceeded by controllable water quality factors. The Plan contains as a "general objective", a reiteration of the SWRCB's Resolution No. 68-16, "Statement of Policy With Respect to Maintaining High Quality Waters in California". That policy provides that whenever existing water quality is better than the numerical objectives established in the Plan, such existing high quality shall be maintained. To implement that Policy, wastewater discharges should be regulated to result in the best practicable treatment or controls to assure that existing high quality conditions are maintained.

12. Compliance with the effluent limitations contained in these Waste Discharge Requirements will assure protection of the high quality waters of the Smith River and adjacent groundwaters.

13. The permittee is presently governed by Waste Discharge Requirements Order No. 89-2, NPDES Permit No. CA0024627 adopted by the RWQCB on January 26, 1989.

14. The action to readopt waste discharge requirements is exempt for the provisions of the California Environmental Quality Act (Public Resources Code Section 21000, et seq.), in accordance with Section 13389 of the California Water Code.

15. The RWQCB has notified the permittee and interested agencies and persons of its intent to prescribe waste discharge requirements for the discharge and has provided them with an opportunity to submit their written comments and recommendations.

16. The RWQCB, in a public meeting, heard and considered all comments pertaining to the discharge.

THEREFORE, IT IS HEREBY ORDERED that Waste Discharge Requirements Order No. 89-2 is rescinded and the permittee, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

A. DISCHARGE PROHIBITIONS:

1. The discharge of waste to land that is not under the control of the discharger is prohibited

2. The discharge of any waste not specifically regulated by this permit is prohibited.

3. The creation of a pollution, contamination, or nuisance, as defined by Section 13050 of the California Water Code (CWC), is prohibited. (Health and Safety Code, Section 5411)

4. The discharge of sludge except as provided for in this order is prohibited.

5. The discharge of untreated waste from anywhere in the collection, treatment, or disposal facility is prohibited.

6. The discharge of waste to the Smith River or its tributaries is prohibited at all times.

B. EFFLUENT LIMITATIONS:

1. Representative samples of the discharge shall not contain constituents in excess of the following limits:

Constituent	Units	30-Day Average[1]	7-Day Average[2]	Daily Maximum
BOD_5	mgl/l lb/day[3]	10 63	15 94	30 283
Suspended Solids	mg/l lb/day	10 63	15 94	30 283
Settleable Solids	ml/l	0.1	---	0.1
Coliform Organisms (total)	MPN/100 ml	2.2[4]	---	23

---

[1]The arithmetic mean of the values for effluent samples collected in a period of 30 consecutive days.

[2]The arithmetic mean of the values for effluent samples collected in a period of 7 days.

[3]The daily discharge (lbs/day) is obtained from the following calculation for any calendar day:

$$\text{Daily Discharge (lb/day)} = \frac{8.34}{N} \sum_{i}^{N} Q_1 C_1$$

in which N is the number of samples analyzed in any calendar day. Q_1 and C_1 ar the flow rate (mgd and the constituent concentration (mg/l), respectively, which are associated with each of the N samples which may be taken in any calendar year. If a composite sample is taken, C_1 is the average flow rate occurring during the period over which samples are composited.

[4]Median

Constituent	Units	30-Day Average[1]	7-Day Average[2]	Daily Maximum
Chlorine Residual	mg/l	---	---	0.1
Hydrogen Ion	pH	Between 6.5 and	8.5	
Grease & Oil	mg/l	15	---	20
Nitrate as N	mg/l	5	---	---
Total Phosphorus	mg/l	2	---	3
Turbidity	NTU	5	---	---

2. The mean daily dry weather flow of waste shall not exceed 0.75 mgd averaged over a period of 30 consecutive days. The peak flow shall not exceed 1.13 mgd.

C. RECEIVING WATER LIMITATIONS:

1. The discharge shall not contain taste or odor producing substances in concentrations that cause nuisance or adversely affect beneficial uses of the receiving waters.

2. The discharge shall not cause the pH of the groundwater to be depressed below 6.5 not raised above 8.5

3. The discharge shall not cause affected groundwaters to exceed the objectives declared in the Basin Plan, nor shall such groundwaters contain concentrations of bacterial indicators or chemical constituents in excess of the limits specified in the California Code of Regulations Title 14 - Drinking Water Standards.

4. The discharge shall not cause esthetically undesirable discoloration of the receiving waters.

5. The discharge shall not contain concentrations of biostimulants which promote objectional aquatic growths to the extent that such growths cause nuisance or adversely affect beneficial uses.

6. The discharge shall not cause groundwater to contain toxic substances in concentrations that are toxic to, or that produce detrimental physiological responses in humans , plant or animal life or adversely affect beneficial uses of groundwater.

7. The discharge shall not cause a measurable temperature change in the receiving waters.

8. The discharge shall not cause the accumulation of pesticides, fungicides, or wood treatment chemical concentrations in soils.

9. The discharge shall not cause a visible film or coating on the surface of receiving waters that cause nuisance or that otherwise adversely affect beneficial uses.

D. SOLIDS DISPOSAL

Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of a legal point of disposal, and in accordance with the provisions of Title 23, Division 3, Chapter 15 of the California Code of Regulations.

E. PROVISIONS:

1. **Availability**

A copy of this Order shall be maintained at the discharge facility and be available at all times to operating personnel.

2. **Severability**

Provisions of these waste discharge requirements are severable. If any provision of these requirements is found invalid, the remainder of these requirements shall not be affected.

3. **Operation and Maintenance**

The discharger must maintain in good working order and operate as efficiently as possible any facility or control system installed by the discharger to achieve compliance with the waste discharge requirements.

4. **Change in Discharge**

The discharger must promptly report to the RWQCB any material change in the character, location, or volume of the discharge.

5. **Change in Ownership**

In the event of any change in control or ownership of land or waste discharge facilities presently owned or controlled by the discharger, the discharger must notify the succeeding owner or operator of the existence of this Order by letter, a copy of which must be forwarded to this office.

6. **Vested Rights**

This Order does not convey any property rights of any sort or any exclusive privileges. The requirements prescribed herein do not authorize the commission of any act causing injury to persons or property, nor protect the discharger from his liability under federal, State, or local laws, nor create a vested right for the discharger to continue the waste discharge.

7. **Monitoring**

The discharger must comply with the Contingency Planning and Notification Requirements Order No. 74-151 and the Monitoring and Reporting Program No. 94-71 and any modifications to these documents as specified by the Executive Officer. Such documents are attached to this Order and incorporated herein. Chemical, bacteriological, and bioassay analyses must be conducted at a laboratory certified for such analyses by the State Department of Health Services. In the event a certified laboratory is not available to the discharger, analyses performed by a noncertified laboratory will be accepted provided:

a. A quality assurance/quality control program is instituted by the laboratory. A manual containing the steps followed in this program must be kept in the laboratory and available for inspection by staff of the RWQCB. The quality assurance/quality control program must conform to EPA guidelines or procedures approved by the RWQCB.
b. The laboratory will become certified within the shortest practicable time if the State Certification Program is resumed.

8. **Inspections**

The discharger shall permit authorized staff of the RWQCB:

a. entry upon premises in which an effluent source is located or in which any required records are kept;
b. access to copy any records required to be kept under terms and conditions of this Order;
c. inspection of monitoring equipment or records; and
d. sampling of any discharge.

9. **Noncompliance**

In the event the discharger is unable to comply with any of the conditions of this Order due to:

a. breakdown of waste treatment equipment;
b. accidents caused by human error or negligence; or
c. other causes such as acts of nature;

the discharger must notify the Executive Officer by telephone as soon as he or his agents have knowledge of the incident and confirm this notification in writing within two weeks of the telephone notification. The written notification shall include pertinent information explaining reasons for the noncompliance and shall indicate the steps taken to correct the problem and the dates thereof, and the steps being taken to prevent the problem from recurring.

10. **Revision of Requirements**

The RWQCB will review this Order periodically and may revise requirements when necessary.

The discharger shall file a report of waste discharge at least 120 days before making any material change or proposed change in the character, location, or volume of the discharge.

Certification

I, Benjamin D. Kor, Executive Officer, do hereby certify that the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, North Coast Region, on June 23, 1994.

Ben Kor

Benjamin D. Kor
Executive Officer

(pelwdr94)

California Regional Water Quality Control Board
North Coast Region

MONITORING AND REPORTING PROGRAM NO. 94-71

FOR THE

CALIFORNIA DEPARTMENT OF CORRECTIONS
PELICAN BAY STATE PRISON

Del Norte County

SEWAGE TREATMENT PLANT MONITORING

The sampling devices currently in use at the Pelican Bay State Prison Sewage Treatment Plant are satisfactory for the composite samples required by this monitoring program. If composite sampling devices are not used, composite grab samples may be substituted. The sampling interval for composite grab samples shall be no more than 1 hour.

Influent Monitoring

Influent samples shall be collected at any point in the facility headworks at which all waste flowing into the plant is present and prior to any treatment. Samples of the influent waste shall be collected on the same day that samples for effluent analyses are collected and analyzed for the following:

Constituent	Units	Sample Type	Frequency
BOD_5 (20°C, 5 day)	mg/l	8 hour composite	Twice Weekly
Suspended Solids	mg/l	8 hour composite	Daily
Flow (Mean and Peak)	mgd	Continuous	Continuous
Total Phosphorous	mg/l	8 hour composite	Monthly

Effluent Monitoring

Effluent samples shall be collected at any point following the Chlorine Contact Chamber and before discharge to the Rapid Infiltration Basins. Samples shall be analyzed for the following:

Constituent	Units	Sample Type	Frequency
Flow (Mean Daily)	mgd	Continuous	Continuous
BOD_5 (20°C, 5 day)	mg/l	24 hour composite	Twice Weekly
Suspended Solids	mg/l	24 hour composite	Daily
Settleable Matter	ml/l	Grab	Weekly
Hydrogen Ion	pH	Grab	Daily
Chlorine Residual[1]	mg/l	Continuous	Daily
Coliform	MPN/100 ml	Grab	Daily
Grease and Oil	mg/l	Grab	Monthly
Turbidity	NTU	Continuous	Continuous
Nitrate Nitrogen	mg/l	24 hour composite	Once Weekly

[1]Following dechlorination

Constituent	Units	Sample Type	Frequency
Total Phosphorus	mg/l	24 hour composite	Twice Weekly
Chloride	mg/l	24 hour composite	Twice Weekly
Halogenated Volatile Organics (EPA Method 601)	mg/l	Grab	Annually
Arsenic	mg/l	24 hour composite	Annually
Cadmium	mg/l	24 hour composite	Annually
Copper	mg/l	24 hour composite	Annually
Chromium (hexavalent)	mg/l	24 hour composite	Annually
Lead	mg/l	24 hour composite	Annually
Mercury	mg/l	24 hour composite	Annually
Nickel	mg/l	24 hour composite	Annually
Silver	mg/l	24 hour composite	Annually
Zinc	mg/l	24 hour composite	Annually
Cyanide	mg/l	24 hour composite	Annually

Groundwater Monitoring

Samples shall be collected from the established network of groundwater monitoring wells which are representative of the upgradient and downgradient conditions of the sludge disposal area and rapid infiltration basins.

Groundwater samples taken from the sludge injection network shall be analyzed for the following:

Constituent	Units	Sample Type	Frequency
Total Dissolved Solids	mg/l	Grab	Monthly
Nitrate Nitrogen	mg/l	Grab	Monthly
Phosphate (Reactive)	mg/l	Grab	Monthly
Chloride	mg/l	Grab	Monthly

Groundwater samples taken from the RIB network shall be analyzed for the following:

Constituent	Units	Sample Type	Frequency
Nitrate Nitrogen	mg/l	Grab	Monthly
Chloride	mg/l	Grab	Monthly

Concurrent with the samples; observe, record, and analyze the variations in water levels within the monitoring wells. Water level measurements shall also be taken at the rapid infiltration basin (RIB) monitoring wells whenever any storm event exceeds 2 inches of precipitation in a 24-hour period. Smith River stage heights shall be determined whenever water levels are measured in the RIB wells. All water level measurements and stage heights shall be referenced to a common datum.

Solids Disposal

Annually, following the completion of sludge disposal for the year. the discharger shall provide a report to the Regional Water Quality Control Board (RWQCB) which shall, as a minimum, include the following:

a. The quantity of sludge disposed (in dry tons and gallons).
b. The quality of sludge disposed (i.e., well stabilized, odor free, metals and nitrogen concentrations)
c. The specific location of the application site(s) where sludge was injected or spread.
d. Analysis of the soil pH prior to sludge disposal.
e. The amount of nitrogen (N) in pounds per acre.

A representative sample treated sludge shall be analyzed by EPA Method 6010 (ICAP SCAN) annually. The results shall be reported on a dry weight basis.

Analytical Methods

Suitable analytical methods are those specified in 40 CFR 136, and Standard Methods for the Examination of Water and Wastewater, 17th Edition, and the 40 CFR Part 503 sludge regulations unless otherwise stated. Any other protocols must be approved by the RWQCB prior to use.

All analytical data must be uncensored with the method detection limits identified. Only data from certified laboratories will be accepted.

Reporting

Monitoring reports shall be submitted to the RWQCB for each month on or before the 15th day of the following month. The annual summary report is due by January 31st of each year. The annual sludge report should be incorporated into the annual summary report.

Ordered By Ben Kor
Benjamin D. Kor
Executive Officer

June 23, 1994

(pelmon94)

California Regional Water Quality Control Board
North Coast Region

CONTINGENCY PLANNING AND NOTIFICATION REQUIREMENTS

FOR

ACCIDENTAL SPILLS AND DISCHARGES

ORDER NO. 74-151

The California Regional Water Quality Control Board, North Coast Region, finds that:

1. Section 13225 of the Porter-Cologne Water Quality Act requires the Regional Board to perform general duties to assure positive water quality control.

2. The Regional Board has been advised of situations in which preparations for, and response to accidental discharges and spills have been inadequate.

3. Persons discharging waste or conveying, supplying, storing, or managing wastes or hazardous materials have the primary responsibility for contingency planning, incident reporting and continuous and diligent action to abate the effects of such unintentional or accidental discharge.

THEREFORE, IT IS HEREBY ORDERED THAT:

I. All persons who discharge wastes or convey, supply, store, or otherwise manage wastes or other hazardous material shall:

   A. Prepare and submit to this Regional Board, according to a time schedule prescribed by the Executive Officer, a contingency plan defining the following:

      1. Potential locations and/or circumstances under which accidental discharge incidents might be expected to occur,

      2. Possible water quality effects of accidental discharges,

      3. The conceptual plan for cleanup and abatement of accidental discharge incidents, including:

         a. The individual who will be in charge of cleanup and abatement activities on behalf of the discharger,

         b. The equipment and manpower available to the discharger to implement the cleanup and abatement plans,

   B. Immediately report to the Regional Board any accidental discharge incidents. Such notification shall be made by telephone as soon as the responsible person or his agent has knowledge of the incident.

   C. Immediately begin diligent and continuous action to cleanup and abate the effects of any unintentional or accidental discharge. Such action shall include temporary measures to abate the discharge prior to completing permanent repairs to damaged facilities.

D. Confirm the telephone notification in writing within two weeks of the telephone notification. The written notification shall include: reasons for the discharge, duration and volume of the discharge, steps taken to correct the problem and steps being taken to prevent the problem from recurring.

II. Upon original receipt of phone report (I.B.), the Executive Officer shall immediately notify all affected agencies and known users of waters affected by the unintentional or accidental discharge.

III. Provide updated information to the Regional Board in the event of change of staff, size of the facility, or change of operating procedures which will affect the previously established contingency plan.

IV. The Executive Officer or his employees shall maintain liaison with the discharger and other affected agencies and persons to provide assistance in cleanup and abatement activities.

V. The Executive Officer shall transmit copies of this Order to all persons whose discharges of waste handling activities are governed by Waste Discharge Requirements or an NDPES permit. Such transmittal shall include a current listing of telephone numbers of the Executive Officer and his key employees to facilitate compliance with Item I.B of this Order.

Ordered by Ben Kor

Benjamin D. Kor
Executive Officer

July 24, 1974
(Retyped February 15, 1990)

Your primary notification should be to the Regional Board office in Santa Rosa at (707) 576-2220. During off hours, you will be able to leave a recorded message at that number and, if you have a spill or discharge emergency, you will also be referred to the State Office of Emergency Services (OES) at (800) 852-7550. OES maintains a roster of key employees and will relay your notification to Regional Board staff.

California Regional Water Quality Control Board
North Coast Region

GENERAL MONITORING AND REPORTING PROVISIONS

February 3, 1971
(Retyped May 20, 1993)

GENERAL PROVISIONS FOR SAMPLING AND ANALYSIS

Unless otherwise noted, all sampling, sample preservation, and analyses shall be conducted in accordance with the current edition of "Standard Methods for the Examination of Water and Waste Water" or approved by the Executive Officer.

All analyses shall be performed in a laboratory certified to perform such analyses by the California State Department of Health or a laboratory approved by the Executive Officer.

All samples shall be representative of the waste discharge under the conditions of peak load.

GENERAL PROVISIONS FOR REPORTING

For every item where the requirements are not met, the discharger shall submit a statement of the actions undertaken or proposed which will bring the discharge in full compliance with requirements at the earliest time and submit a timetable for correction.

By January 30 of each year, the discharger shall submit an annual report to the Regional Board. The report shall contain both tabular and graphical summaries of the monitoring data obtained during the previous year. In addition, the discharger shall discuss the compliance record and the corrective actions taken or planned which may be needed to bring the discharge into full compliance with the waste discharge requirements.

The discharger shall file a written report within 90 days after the average dry weather flow for any month that equals or exceeds 75 percent of the design capacity of the waste treatment or disposal facilities. The report shall contain a schedule for studies, design, and other steps needed to provide additional capacity or limit the flow below the design capacity prior to the time when the waste flow rate equals the capacity of the present units.